

## Type II transformation

Given joint PDF  $f_{XY}(x, y)$  or CDF  $F_{XY}(x, y)$  and  $S_{X,Y}$  of  $X$  and  $Y$   
find joint PDF  $f_{WZ}(w, z)$  (or CDF  $F_{WZ}(w, z)$ ) of the two RVs

$$\boxed{W = g(X, Y)} \quad \text{and} \quad \boxed{Z = h(X, Y)}$$

## Type II transformation with unique solution

Given  $W = g(X, Y)$  and  $Z = h(X, Y)$

if unique solutions for  $x$  and  $y$  in terms of  $w$  and  $z$ ,  $x = g^{-1}(w, z)$  and  $y = h^{-1}(w, z)$ , then

Joint CDF of  $W$  and  $Z$

$$F_{WZ}(w, z) = F_{XY}\left(g^{-1}(w, z), h^{-1}(w, z)\right)$$

and joint PDF

$$f_{WZ}(w, z) = \frac{\partial^2}{\partial w \partial z} F_{WZ}(w, z) = f_{XY}\left(g^{-1}(w, z), h^{-1}(w, z)\right) |\det(J)|$$

where matrix  $J$  is the Jacobian matrix of transformation from  $(x, y)$  to  $(w, z)$

$$J = \frac{\partial(x, y)}{\partial(w, z)} = \begin{bmatrix} \frac{\partial x}{\partial w} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial w} & \frac{\partial y}{\partial z} \end{bmatrix} \quad \text{and} \quad \det(J) = |J| = \begin{vmatrix} \frac{\partial x}{\partial w} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial w} & \frac{\partial y}{\partial z} \end{vmatrix} = 1 / \begin{vmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \\ \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{vmatrix}$$

### Example: Linear Transformation [Kay, Example 12.6 for jointly Gaussian RVs]

Let  $W = g(X, Y) = aX + bY$  and  $Z = h(X, Y) = cX + dY$ , with  $a, b, c, d \in \mathbb{R}$  deterministic

In vector form,

$$\begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{where } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

If matrix  $A$  is invertible, then

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} w \\ z \end{bmatrix}, \quad \text{where } A^{-1} = \frac{1}{(ad - bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

that yields

$$x = g^{-1}(w, z) = \frac{1}{(ad - bc)}(dw - bz) \quad \text{and} \quad y = h^{-1}(w, z) = \frac{1}{(ad - bc)}(-cw + az)$$

$\Rightarrow$  joint CDF of  $W$  and  $Z$ ,

$$F_{WZ}(w, z) = F_{XY}\left(\frac{1}{(ad - bc)}(dw - bz), \frac{1}{(ad - bc)}(-cw + az)\right)$$

For joint PDF, find magnitude of determinant of Jacobian matrix of transformation  $J$ ,

$$\begin{aligned}
 |\det(J)| &= \begin{vmatrix} \frac{\partial x}{\partial w} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial w} & \frac{\partial y}{\partial z} \end{vmatrix} = \begin{vmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{vmatrix} \\
 &= \left| \frac{ad}{(ad-bc)^2} - \frac{bc}{(ad-bc)^2} \right| = \left| \frac{(ad-bc)}{(ad-bc)^2} \right| = \frac{1}{|ad-bc|} = |\det(A^{-1})|
 \end{aligned}$$

For linear transformation,  $\boxed{|\det(J)| = |\det(A^{-1})|}$

Substituting in the general expression yields

$$f_{WZ}(w, z) = f_{XY}\left(g^{-1}(w, z), h^{-1}(w, z)\right) |\det(A^{-1})|$$

results in

$$\boxed{f_{WZ}(w, z) = \frac{1}{|ad-bc|} f_{XY}\left(\frac{1}{(ad-bc)}(dw - bz), \frac{1}{(ad-bc)}(-cw + az)\right)}$$

Example: Rayleigh fading communications channel with (pages 5-7)

- ▶ transmit signal:  $s(t) = \cos(2\pi f_0 t)$ , deterministic
- ▶ received (noiseless) signal:  $z(t) = R \cos(2\pi f_0 t + \Phi)$ , random with
  - $R$  is Rayleigh distributed (amplitude fading)
  - $\Phi$  is uniformly distributed from  $-\pi$  to  $\pi$  (phase fading)

To simulate the two Rayleigh fading random variables, transform two Gaussian RVs that are independent and identically distributed (iid),  $X \sim \mathcal{N}(0, \sigma^2)$ ,  $Y \sim \mathcal{N}(0, \sigma^2)$  with joint Gaussian PDF

$$\begin{aligned} f_{XY}(x, y) &= \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(x^2 + y^2)\right) \\ &= \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}x^2\right) \exp\left(-\frac{1}{2\sigma^2}y^2\right) = f_X(x) f_Y(y) \end{aligned}$$

using transformation

$$R = g(X, Y) = \sqrt{X^2 + Y^2} \text{ and } \Phi = h(X, Y) = \arctan(Y/X)$$

To demonstrate, solve for  $x$  and  $y$  in terms of  $r = \sqrt{x^2 + y^2}$  and  $\phi = \arctan(y/x)$  yields one unique solution

$$x = r \cos(\phi) \text{ and } y = r \sin(\phi)$$

The joint PDF of  $R$  and  $\Phi$  can be found as

$$f_{R\Phi}(r, \phi) = f_{X,Y}(r \cos(\phi), r \sin(\phi)) |\det(J)|$$

where

$$|\det(J)| = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \cos(\phi) & -r \sin(\phi) \\ \sin(\phi) & r \cos(\phi) \end{vmatrix} = |r \cos^2(\phi) + r \sin^2(\phi)| = r > 0$$

Thus,

$$f_{R\Phi}(r, \phi) = r f_{X,Y}(r \cos(\phi), r \sin(\phi)), \quad r > 0, -\pi < \phi < \pi$$

Note: since  $X$  and  $Y$  are independent, then  $R$  and  $\Phi$  are also independent

To demonstrate, substitute Gaussian  $f_{X,Y}(x, y)$  in  $f_{R\Phi}(r, \phi)$  and use  $\cos^2(\phi) + \sin^2(\phi) = 1$ ,

$$\begin{aligned} f_{R\Phi}(r, \phi) &= r f_{X,Y}(r \cos(\phi), r \sin(\phi)) = r f_X(r \cos(\phi)) f_Y(r \sin(\phi)) \\ &= \frac{r}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(r^2 \cos^2(\phi) + r^2 \sin^2(\phi))\right) \\ &= \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) \\ &= \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) \frac{1}{2\pi} \\ &= f_R(r) f_\Phi(\phi) \end{aligned}$$

where

$f_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right)$  is the Rayleigh distribution of  $R$  with parameter  $\sigma^2$

$f_\Phi(\phi) = \frac{1}{2\pi}$  is the uniform distribution of  $\Phi$  from  $-\pi$  to  $\pi$